

```

import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt

data=pd.read_csv('salary_data.csv')

data.head()

```

	YearsExperience	Salary
0	1.1	39343.0
1	1.3	46205.0
2	1.5	37731.0
3	2.0	43525.0
4	2.2	39891.0

```

data.shape

(30, 2)

data.isnull().sum()

YearsExperience    0
Salary            0
dtype: int64

x=data.iloc[:,1].values
y=data.iloc[:,2].values

x_train, x_test, y_train, y_test = train_test_split(x,y,test_size=0.2,
random_state=42)

model=LinearRegression()
model.fit(x_train, y_train)
y_pred=model.predict(x_test)

train_accuracy = model.score(x_train, y_train)
test_accuracy = model.score(x_test, y_test)

print("Training Accuracy (R² Score):", train_accuracy)
print("Testing Accuracy (R² Score):", test_accuracy)

Training Accuracy (R² Score): 0.9645401573418146
Testing Accuracy (R² Score): 0.9024461774180497

print(y_pred)
print(y_test)

[[115790.21011287]
 [ 71498.27809463]
 [102596.86866063]
 [ 75267.80422384]]

```

```
[ 55477.79204548]
[ 60189.69970699]]
[[112635.]
 [ 67938.]
 [113812.]
 [ 83088.]
 [ 64445.]
 [ 57189.]]
```

```
plt.scatter(x_train, y_train, color='blue')
plt.plot(x_train, model.predict(x_train), color='red')
plt.title('SALARY VS EXPERIENCE (training set)')
plt.xlabel('Experience in Years')
plt.ylabel('Salary in Rupees')
plt.show()
```



```
plt.scatter(x_test, y_test, color='blue')
plt.plot(x_train, model.predict(x_train), color='red')
plt.title('SALARY VS EXPERIENCE (testing set)')
plt.xlabel('Experience in Years')
plt.ylabel('Salary in Rupees')
plt.show()
```



```
import pandas as pd
data = {
    'Weight': [79, 69, 73, 95, 82, 55, 69, 71, 64, 69],
    'Height': [1.80, 1.68, 1.82, 1.70, 1.87, 1.55, 1.50, 1.78, 1.67,
1.64],
    'Age': [35, 39, 25, 60, 27, 18, 89, 42, 16, 52],
    'Gender': ['Male', 'Male', 'Male', 'Male', 'Male',
    'Female', 'Female', 'Female', 'Female', 'Female']
}
df = pd.DataFrame(data)
df.to_csv('body_weight.csv', index=False)
print("body_weight.csv created successfully!")
print(df)
```

body_weight.csv created successfully!

	Weight	Height	Age	Gender
0	79	1.80	35	Male
1	69	1.68	39	Male
2	73	1.82	25	Male
3	95	1.70	60	Male
4	82	1.87	27	Male
5	55	1.55	18	Female
6	69	1.50	89	Female
7	71	1.78	42	Female
8	64	1.67	16	Female
9	69	1.64	52	Female

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
```

```
data = pd.read_csv('body_weight.csv')
```

```
data.head()
```

	Weight	Height	Age	Gender
0	79	1.80	35	Male
1	69	1.68	39	Male
2	73	1.82	25	Male
3	95	1.70	60	Male
4	82	1.87	27	Male

```
data.shape
```

```
data.isnull().sum()
```

```
Weight      0
Height      0
Age         0
Gender      0
dtype: int64
```

```
data['Gender'] = data['Gender'].map({'Male':1, 'Female':0})
```

```
x = data[['Height', 'Age', 'Gender']]
y = data[['Weight']]

x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(x_train, y_train)

LinearRegression()

y_pred = model.predict(x_test)
print("Predicted Weight:")
print(y_pred)
print("\nActual Weight:")
print(y_test.values)

Predicted Weight:
[[56.85651025]
 [79.10576115]]

Actual Weight:
[[64]
 [69]]

train_accuracy = model.score(x_train, y_train)
test_accuracy = model.score(x_test, y_test)
print("Training Accuracy:", train_accuracy)
print("Testing Accuracy:", test_accuracy)

Training Accuracy: 0.8312942630906153
Testing Accuracy: -11.252468343750815
```